Using AI to make a mobile app to detect head lice nits

Parasite ID, LLC

TEAM

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Inventor of the Smart Pill and water Bottle, Clinical Hospital Pharmacist

Lice DETECTOR

PROBLEM

There is currently no way to **scientifically** determine the presence of an **active** loess Lice Infestation which affects 12 million/yr

Lice infections are frequently

Misdiagnosed 42% of the time



If funding **cannot** be obtained to develop a scientific method to detect head lice. Mankind will be picking through hair looking for lice for decades to come.

The Reasons Head Lice are Misdiagnosed

1. Head lice are fast moving and are overlooked and avoid light
2. Head Lice are small
adult female 2.5-3.3 mm
adult male 2.1-2.6 mm
juvenile .8 mm
3. Head lice nits (eggs) are tiny ovals measuring about 0.8 mm long and 0.3 mm wide. They are frequently mistaken for dandruff.

MARKET

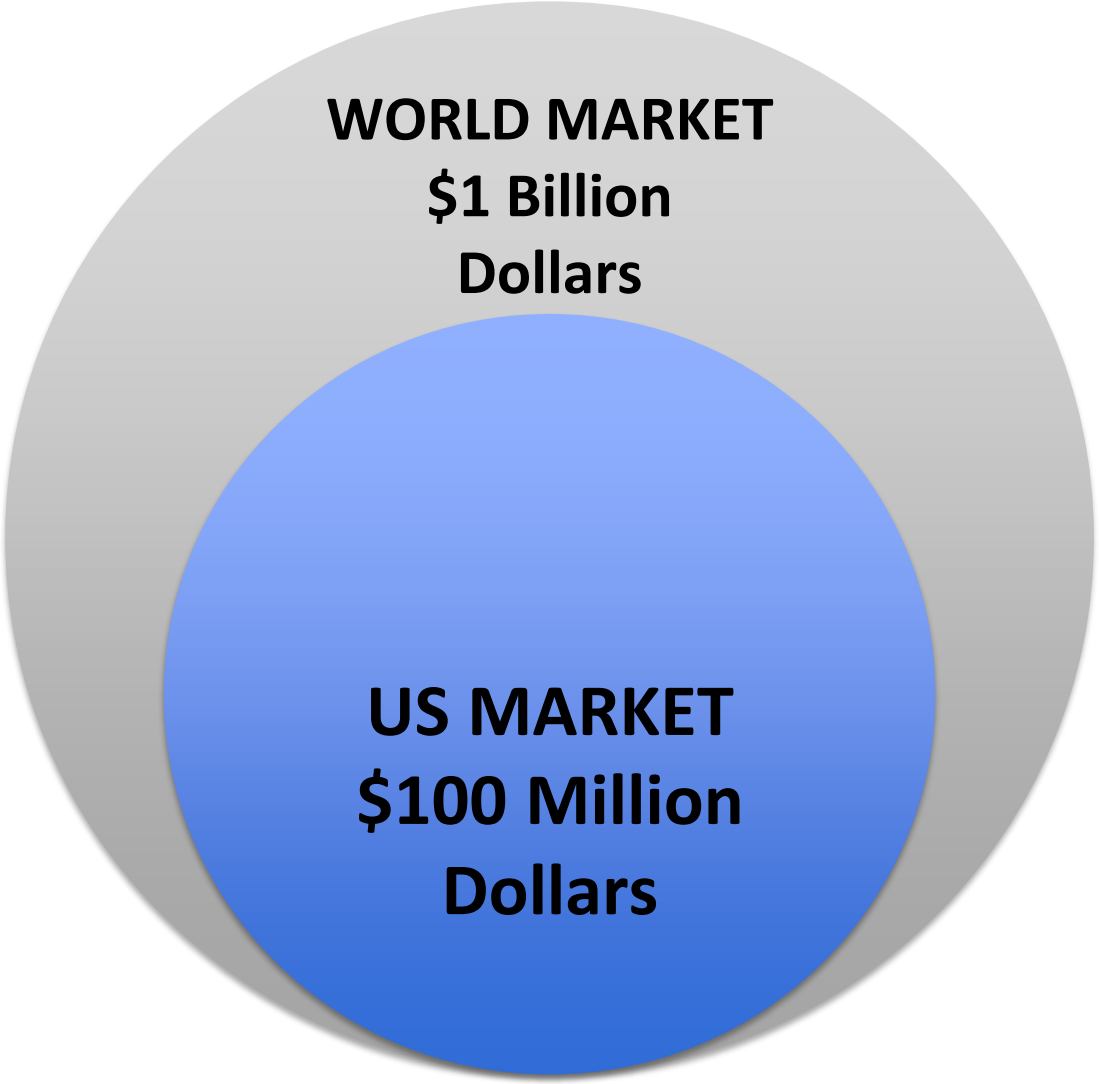
$10 dollars sell price

US Market: 10 Million people with lice

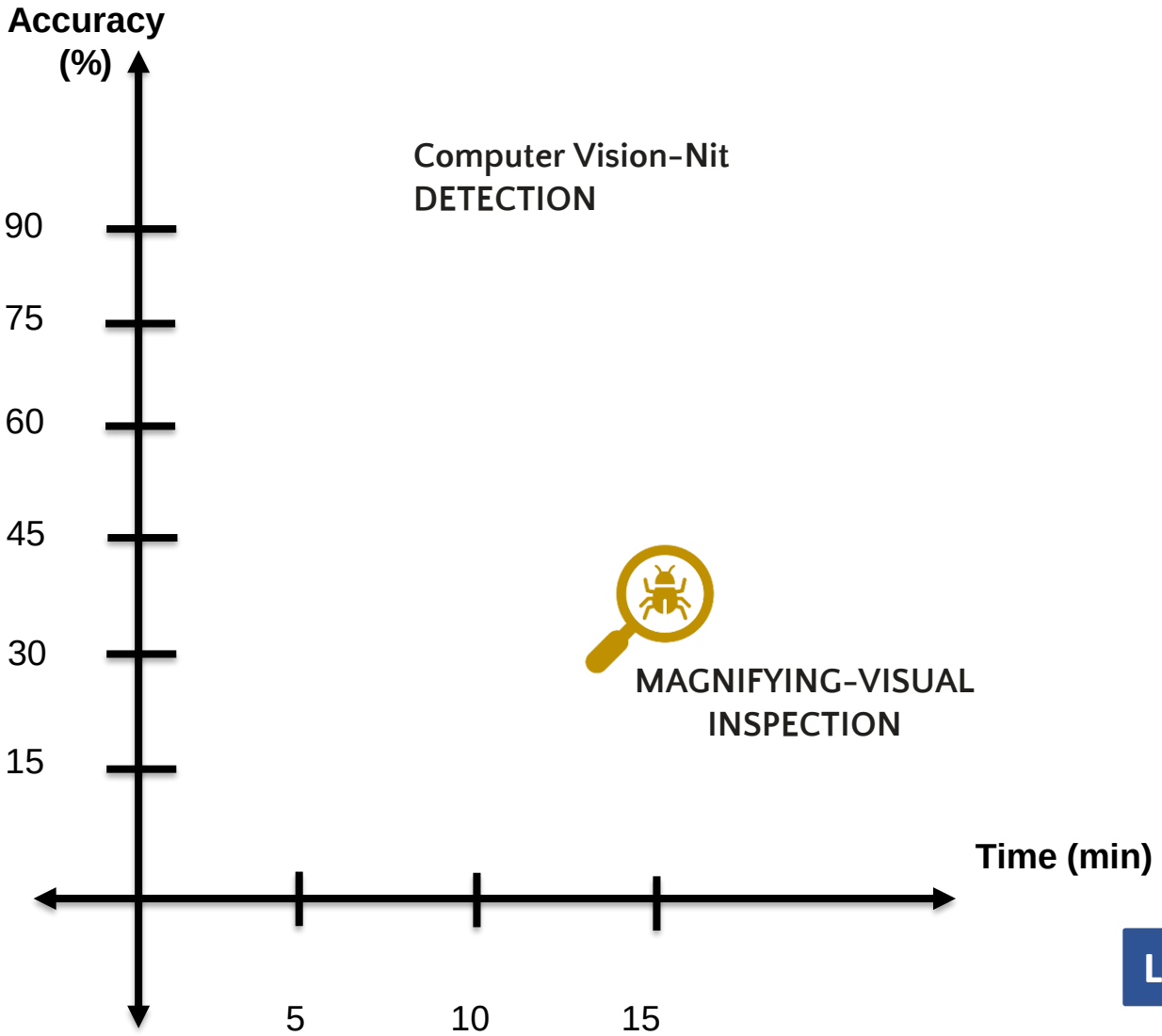
Total addressable market $100 million dollars

World Market: $100 million people with lice

Total addressable market $1 billion dollars.



COMPETITION



Scientifically determine the presence of lice infestations

LICE DETECTOR

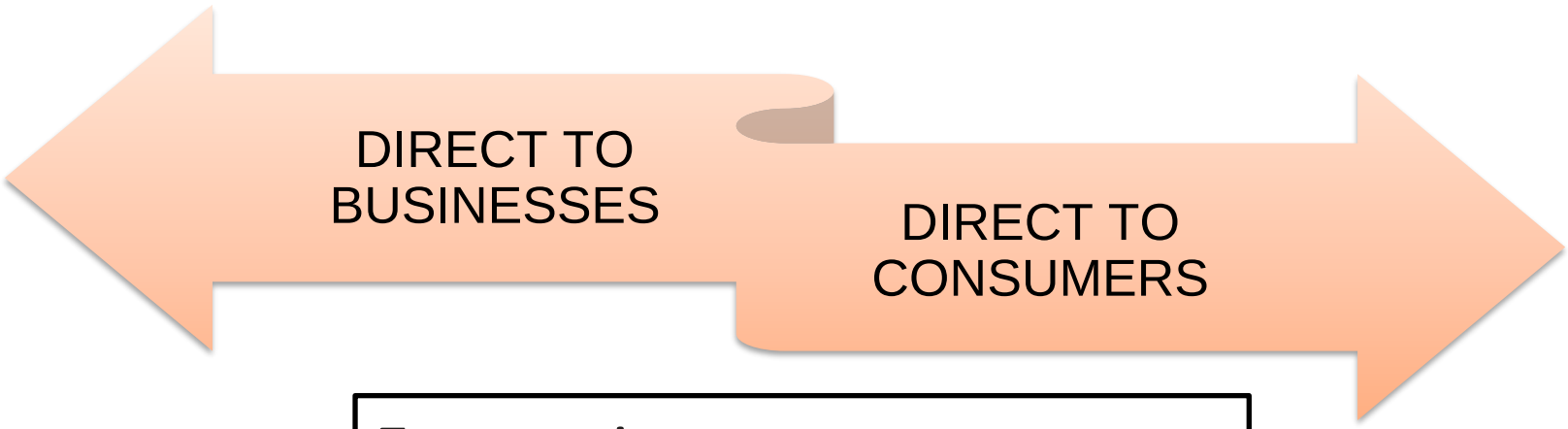
- High Accuracy 95%
- Low amount of Time 5 minutes

VISUAL INSPECTION

- Low Accuracy 32%
-High amount of time 15 minutes

LOW TIME TO DIAGNOSE + HIGH ACCURACY= COST SAVINGS

BUSINESS MODEL



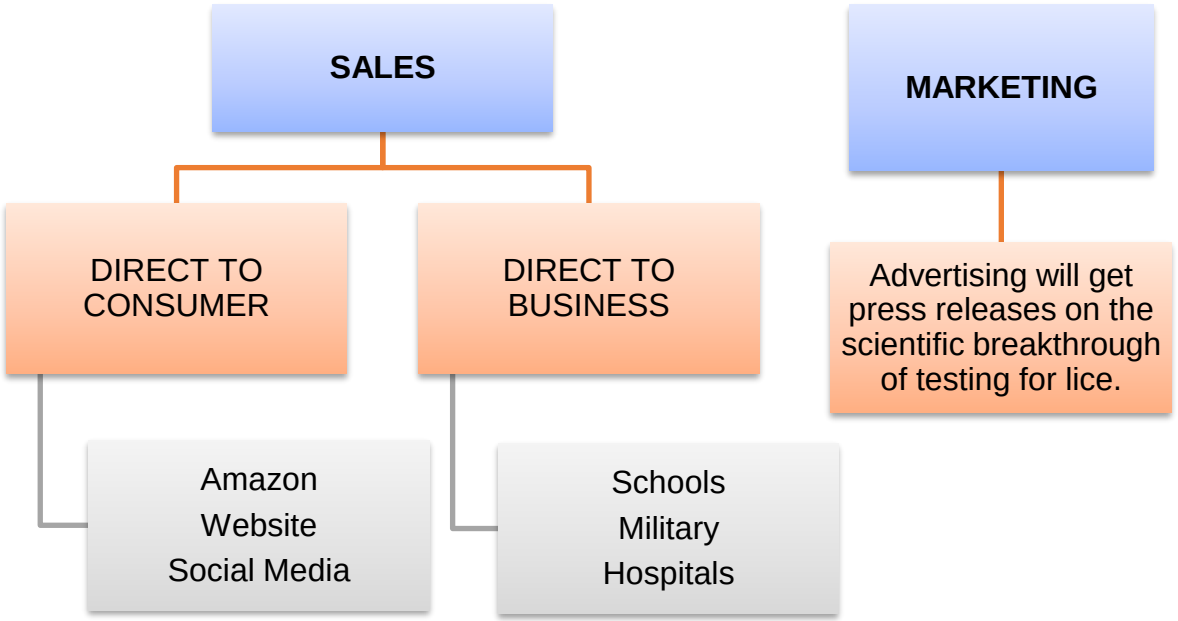
Economics

Estimated sales Price $10
Estimated cost $ 5

Margin 50%

SALES AND MARKETING

- Market Research Completed by a **professional** market research company PAR found that **67%** of persons would purchase a lice test strip if available and they suspected lice.



Developing a mobile app to detect head lice Nits using computer Vision

1. **Software to be Developed on a phone prefer both IOS and android phone.**
2. **Will use phone video to detect head lice nits**
3. **Videos and pics on the head and comb are available on the internet**
4. **Testing will occur when the solution is complete**
5. **https://medlineplus.gov/ency/imagepages/2437.htm**

On Phone image capture and analysis

Smart phone camera will be uses take pictures of head and lice comb

And a neural net training will occur using head lice nits available on the internet

Train the software to recognize head lice nits.

Images are available on the internet such as https://skinsight.com/skin-conditions/pediculosis-capitis-head-lice/

App development

A mobile app will be downloaded to a smart phone will be

Recording keeping will occur on the phone

Payment will occur with each use

Expected Software Environments

Iphone (ios)

Android

AI frameworks to be determined by the students